EXPERIMENT-05

ELECTROMAGNETIC SIMULATION AND S-PARAMETER CHARACTERIZATION OF A MICROWAVE DIRECTIONAL COUPLER AND TWO-WAY POWER DIVIDER

PROGRAM

```
clc;

clear;

close all;

% Electromagnetic Simulation and S-Parameter Characterization

% Microwave Directional Coupler and Two-Way Power Divider

%% Simulation Parameters

f0 = 10e9;

BW = 2e9;

f = linspace(9e9,11e9,201);

x = (f-f0)/(BW/2);

Z0 = 50;

c = 3e8;

lambda0 = c/f0;

L = lambda0/4;

%% 1. Directional Coupler

coupling_dB = -10 + cos(pi*x);

k = 10.^(coupling_dB/20);

through = sqrt(1-k.^2);

beta = 2*pi*f/c;

S11_c = 0.02*ones(size(f));
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S21_c = through.*exp(-1j*beta*L);

S31_c = k.*exp(-1j*(beta*L+pi/2));

S41_c = 10^(-35/20)*ones(size(f));

S11c_dB = 20*log10(abs(S11_c));

S21c_dB = 20*log10(abs(S21_c));

S31c_dB = 20*log10(abs(S31_c));

S41c_dB = 20*log10(abs(S41_c));

%% 2. Two-Way Power Divider

S11_d = 0.02 + 0.16*abs(x);

S21_d = (1/sqrt(2))*(1-0.04*x.^2);

S31_d = (1/sqrt(2))*(1-0.04*x.^2);

S23_d = 0.01 + 0.14*abs(x);

S11d_dB = 20*log10(abs(S11_d));

S21d_dB = 20*log10(abs(S21_d));

S31d_dB = 20*log10(abs(S31_d));

S23d_dB = 20*log10(abs(S23_d));

%% 3. Numerical Results

[~,idx] = min(abs(f-f0));

fprintf('\nMICROWAVE DEVICE SIMULATION RESULTS\n');

fprintf('--------------------------------------\n');

fprintf('Characteristic Impedance = %.1f Ohms\n',Z0);

fprintf('Center Frequency = %.2f GHz\n',f(idx)/1e9);

fprintf('Wavelength = %.2f mm\n',lambda0*1000);

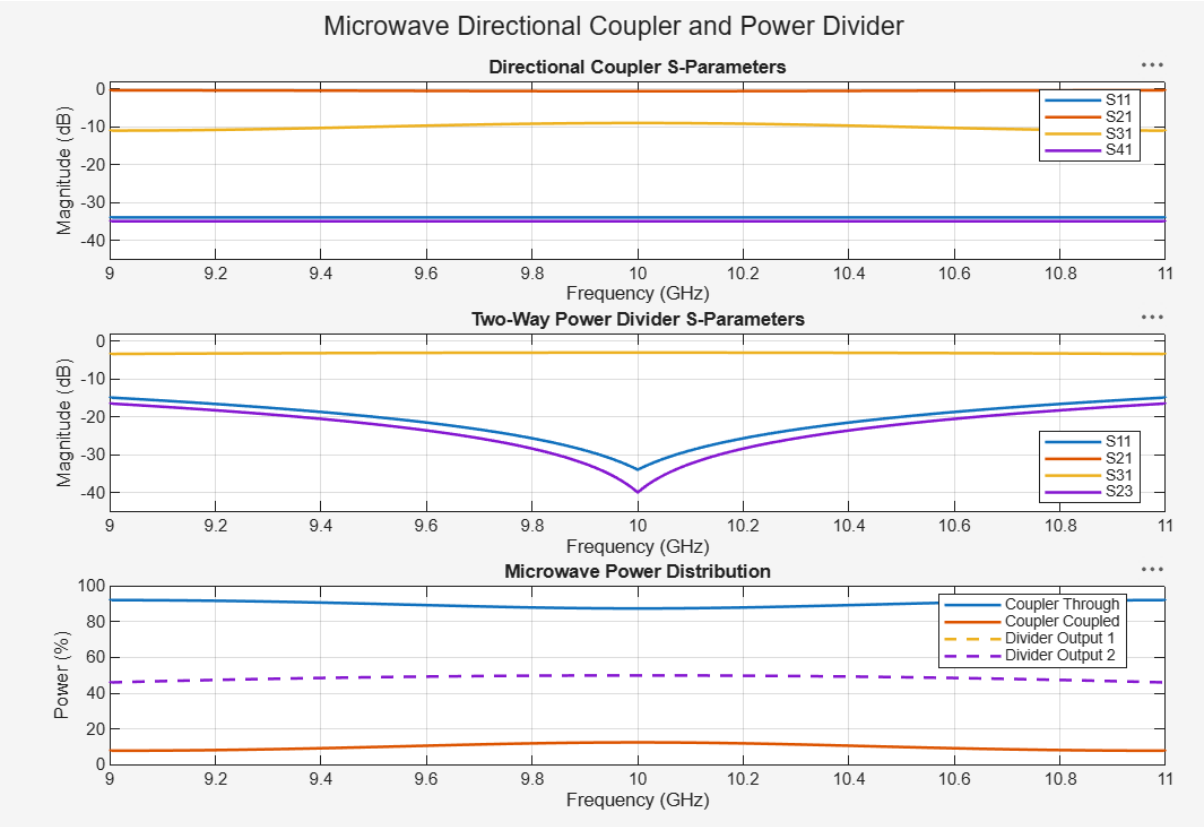
fprintf('Quarter-Wave Length = %.2f mm\n',L*1000);
```

```
fprintf('\nDIRECTIONAL COUPLER\n');
fprintf('S11 = %.2f dB\n',S11c_dB(idx));
fprintf('S21 = %.2f dB\n',S21c_dB(idx));
fprintf('S31 = %.2f dB\n',S31c_dB(idx));
fprintf('S41 = %.2f dB\n',S41c_dB(idx));
fprintf('\nTWO-WAY POWER DIVIDER\n');
fprintf('S11 = %.2f dB\n',S11d_dB(idx));
fprintf('S21 = %.2f dB\n',S21d_dB(idx));
fprintf('S31 = %.2f dB\n',S31d_dB(idx));
fprintf('S23 = %.2f dB\n',S23d_dB(idx));
%% 4. Combined Subplot Figure
figure('Name','Microwave S-Parameter Characterization',...
    'NumberTitle','off');
% Subplot 1: Directional Coupler
subplot(3,1,1);
plot(f/1e9,S11c_dB,'LineWidth',1.5);
hold on;
plot(f/1e9,S21c_dB,'LineWidth',1.5);
plot(f/1e9,S31c_dB,'LineWidth',1.5);
plot(f/1e9,S41c_dB,'LineWidth',1.5);
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Directional Coupler S-Parameters');
```

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legend('S11','S21','S31','S41','Location','best');
ylim([-45 2]);
% Subplot 2: Two-Way Power Divider
subplot(3,1,2);
plot(f/1e9,S11d_dB,'LineWidth',1.5);
hold on;
plot(f/1e9,S21d_dB,'LineWidth',1.5);
plot(f/1e9,S31d_dB,'LineWidth',1.5);
plot(f/1e9,S23d_dB,'LineWidth',1.5);
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Two-Way Power Divider S-Parameters');
legend('S11','S21','S31','S23','Location','best');
ylim([-45 2]);
% Subplot 3: Power Distribution
subplot(3,1,3);
plot(f/1e9,abs(S21_c).^2*100,'LineWidth',1.5);
hold on;
plot(f/1e9,abs(S31_c).^2*100,'LineWidth',1.5);
plot(f/1e9,abs(S21_d).^2*100,'--','LineWidth',1.5);
plot(f/1e9,abs(S31_d).^2*100,'--','LineWidth',1.5);
grid on;
xlabel('Frequency (GHz)');
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ylabel('Power (%)');
title('Microwave Power Distribution');
legend('Coupler Through','Coupler Coupled',...
    'Divider Output 1','Divider Output 2',...
    'Location','best');
sgtitle('Microwave Directional Coupler and Power Divider');
```

OUTPUT



RESULT

The simulation plots show the frequency-dependent behavior of the microwave directional coupler and two-way power divider over 9–11 GHz. These are illustrative behavioral models, not full electromagnetic simulations.